

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

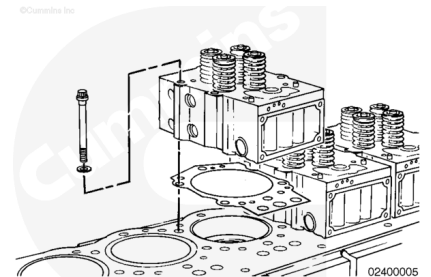
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid contact of hot oil with your skin.



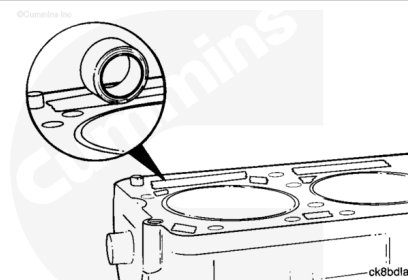
Note : Mechanically actuated injector engines were originally produced with articulated (two-piece) pistons. During 2005 the articulated piston was made obsolete by a one-piece cast iron piston. The articulated pistons are no longer available for production or service. The cast iron piston is backward compatible with the articulated piston for single cylinder replacements, full engine exchanges, or any combination in between, except for mechanically actuated injector engines that have low temperature aftercooling (LTA). Because of an increase in weight with LTA pistons, intermixing is not allowed. If an articulated piston must be replaced on an LTA engine, all six pistons must be replaced with the cast iron piston.

- Disconnect the batteries. See equipment manufacturer service information.

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html>)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-046-tr.html>)
- Drain the oil. Remove the oil sump. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)
- Remove the oil pan adapter cover plate. Refer to Procedure 007-026 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-026-tr.html>)
- Remove the oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-035-tr.html>)

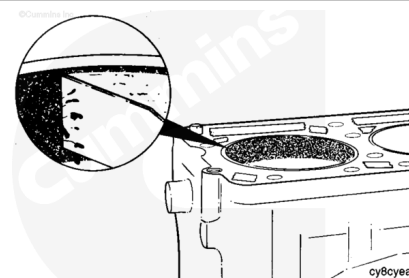
Remove

Protect the push rod galleys, coolant passages, and oil passages from contamination.



⚠ CAUTION ⚠

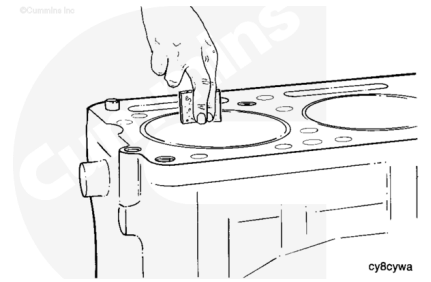
Do not use abrasive paper to remove the carbon deposits. Small particles of abrasive paper will cause severe engine damage.



Use a scraper or a similar blunt-edged tool to loosen the carbon deposits.

⚠ CAUTION ⚠

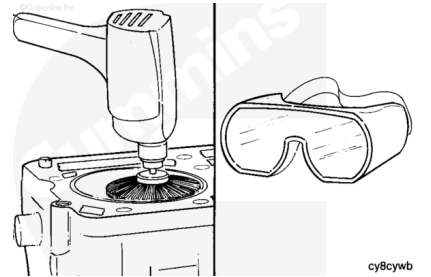
Use extreme care to be sure that no abrasive cleaners or materials are used in the piston ring travel area. Severe engine damage can result.



Remove the remaining carbon with a nylon abrasive pad, Scotch-Brite™ 7448, or equivalent, and solvent. The carbon **must** be removed, but the surface does **not** have to appear like new metal.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using a wire wheel. Flying debris and dirt can cause serious personal injury.



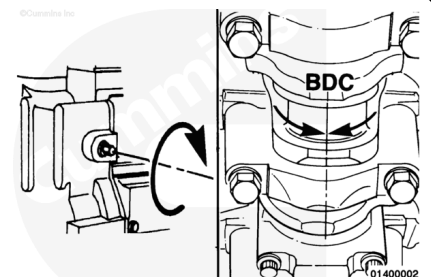
⚠ CAUTION ⚠

Do not use a steel wire wheel of inferior quality. The inferior wire wheel will lose steel bristles during operation and cause additional contamination.

An alternative method to remove the carbon ridge is to use a high quality-steel wire wheel installed in a drill.

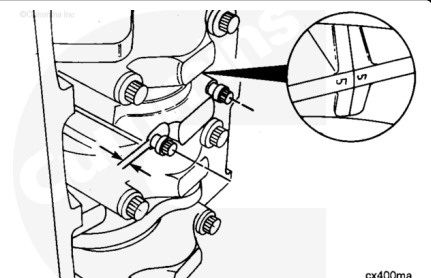
Note : Do not use a steel wire wheel in the piston ring travel area. Operate the wire wheel in a circular motion to remove the deposits.

Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at bottom dead center.



Connecting rods **must** have the cylinder number marked on both the rod and the cap on the side positioned toward the camshaft. Check the rods for correct markings. Use a steel stamp and mark any rod that is **not** correctly marked.

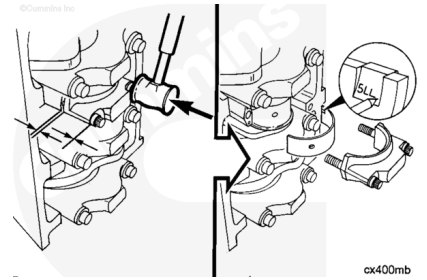
Loosen the capscrews until there is 6 mm [1/4 in] of clearance between the rod cap and the capscrew head.



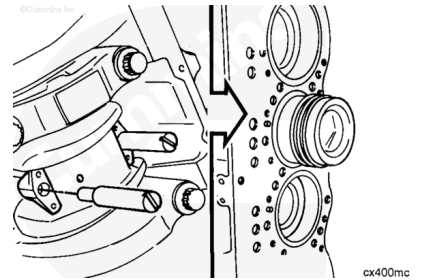
Use a mallet. Tap the connecting rod capscrews until the connecting rod cap and rod separate. Remove the capscrews and the cap.

Remove the lower rod bearing. Use an awl to mark the bearing position in the tang area.

Mark the bearing for future identification or for possible analysis.



Install two connecting rod guide pins, Part Number 3375098, or equivalent, into the connecting rod. Push the piston and connecting rod up until the piston rings are above the cylinder liner.



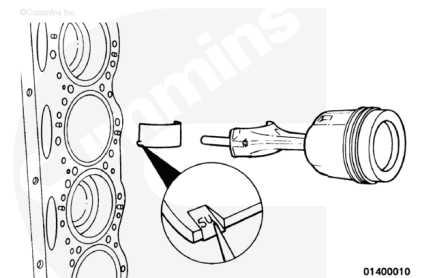
⚠ CAUTION ⚠

Place the piston and rod assembly in a rack to prevent damage to the piston and rod assembly.

Remove the piston and rod assembly.

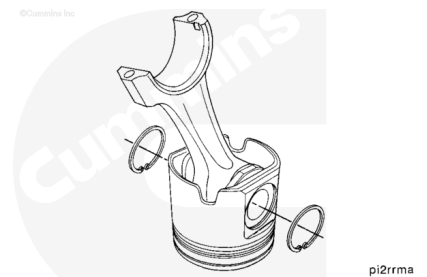
Remove the upper connecting rod bearing.

Use an awl to mark the bearing position in the tang area.



Disassemble

Note : QSK19 mechanically actuated injector engines were originally produced with articulated (two-piece) pistons. During 2005 the articulated piston was made obsolete by a one-piece cast iron piston. The articulated pistons are no longer available for production or service. The cast iron piston is backward compatible with the articulated piston for single cylinder replacements, full engine exchanges, or any combination in between, except for mechanically actuated injector engines that have low temperature aftercooling (LTA). Because of an increase in weight with LTA pistons, intermixing is not allowed. If an articulated piston must be replaced on an LTA engine,



all six pistons must be replaced with the cast iron piston.

Remove piston rings.

Refer to Procedure 001-047 in Section 1
(/qs3/pubsys2/xml/en/procedures/20/20-001-047.html).

Use internal snap ring pliers. Remove the snap rings from both sides of the piston.

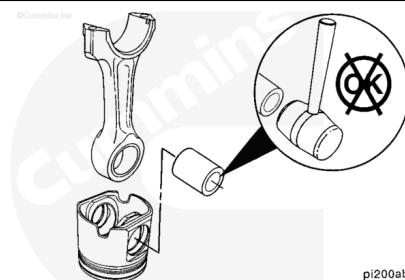
⚠ CAUTION ⚠

Do not use a hammer to remove the piston pin. The piston can distort, causing it to seize in the cylinder liner.

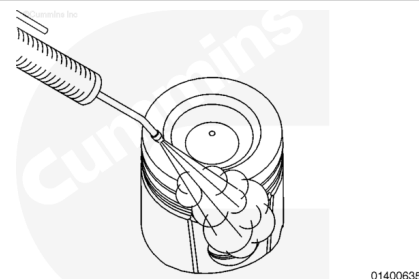
⚠ CAUTION ⚠

Do not drop the piston while removing the piston pin. Personal injury and damage to the piston can result.

Remove the piston pin and separate the connecting rod and piston.



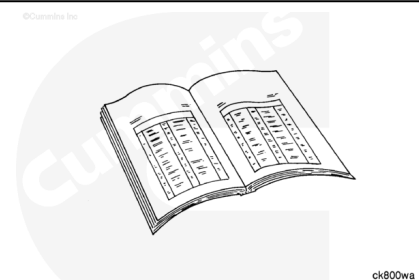
Clean and inspect the pistons. Refer to Procedure 001-043 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-043.html)



Clean and Inspect for Reuse

Clean and inspect the connecting rod for reuse. Refer to Procedure 001-014 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-014-tr.html)

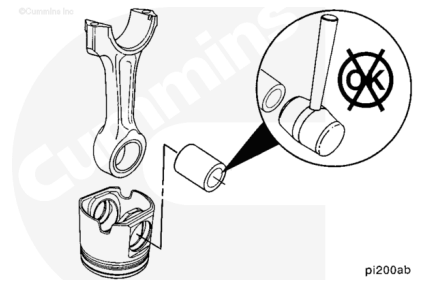
Clean and inspect the pistons for reuse. Refer to Procedure 001-043 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-043.html)



Assemble

⚠ CAUTION ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing it to seize in the cylinder liner.



Install a new snap ring into one of the piston bores.

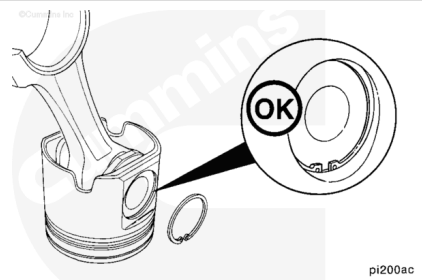
The snap ring **must** be seated completely in the piston groove to prevent engine damage during engine operation.

Use clean engine oil and lubricate the piston pin and connecting rod bushing.

Align the pin bore of the connecting rod with the pin bore of the piston skirt and install the piston pin.

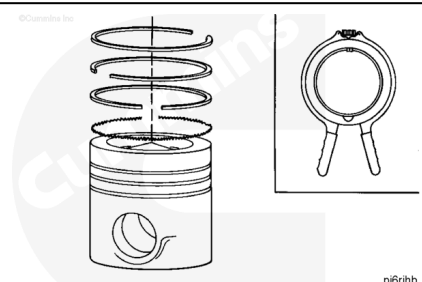
Install a new snap ring in the piston pin bore.

The snap ring **must** be seated completely in the piston groove to prevent engine damage during engine operation.



Install the piston rings.

Refer to Procedure 001-047 in Section 1 ([/qs3/pubsys2/xml/en/procedures/20/20-001-047.html](http://qs3/pubsys2/xml/en/procedures/20/20-001-047.html)).



Install

⚠ CAUTION ⚠

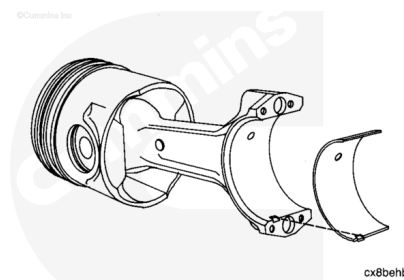
To reduce the possibility of engine damage, do not lubricate the back side of the bearing shells.

Clean the connecting rod and the bearing shells with a lint-free cloth.

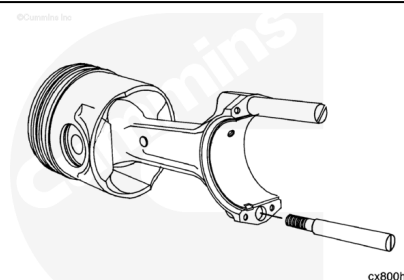
Install the rod bearing. Be sure the tang is positioned as shown. The end of the bearing **must** be even with the cap mounting surface.

Use clean engine oil. Lubricate the bearing surface. The bearings **must** be installed in their original locations if new bearings are **not** used.

Note : All of the rod bearings are identical.

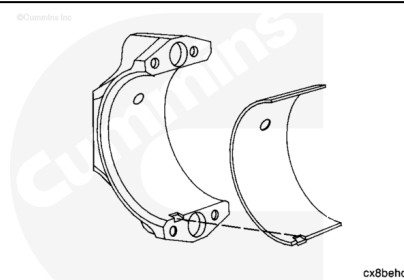


Install two connecting rod guide pins, Part Number 3375098, in the connecting rod. The guide pins will aid the assembly process and protect the crankshaft.



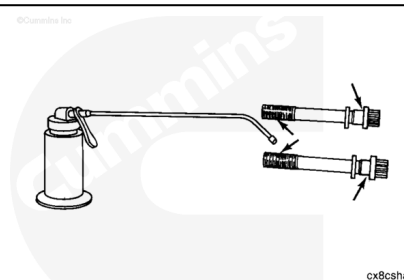
⚠ CAUTION ⚠

The connecting rods and connecting rod caps are not interchangeable. The connecting rods and the connecting rod caps are machined as an assembly. Engine damage will result if they are mixed.

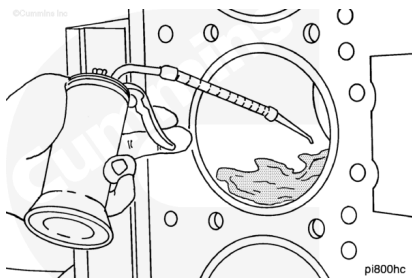


Install the lower bearing shell in the rod cap. Be sure the tang of the bearing shell is in the slot of the cap and the end of the bearing is even with the cap surface.

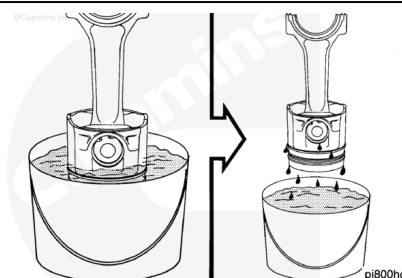
Use clean engine oil. Lubricate the connecting rod capscrews and washers as shown. Install the washers and capscrews in the caps.



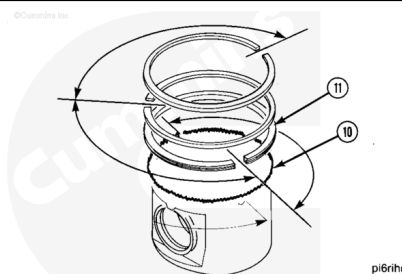
Lubricate the cylinder liner with clean engine oil. The entire bore **must** be lubricated.



Immerse the piston in clean engine oil until the rings are covered. Allow the excess oil to drip off the assembly.



Check to be sure the ring gap position is still correct.

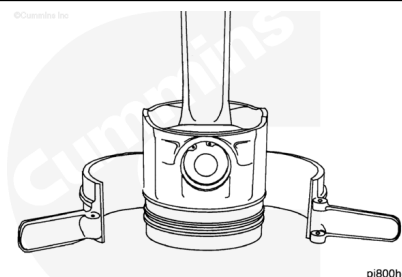


⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the piston rings fit correctly in the piston.

Use a piston ring compressor, Part Number 4918898, or equivalent. Install the ring compressor on the piston.

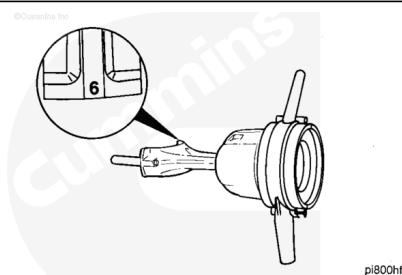
Note : The ring compressor has a tapered bore. The small end of the taper must be positioned toward the piston skirt.



The cylinder number on the connecting rod and connecting rod cap **must** be the same.

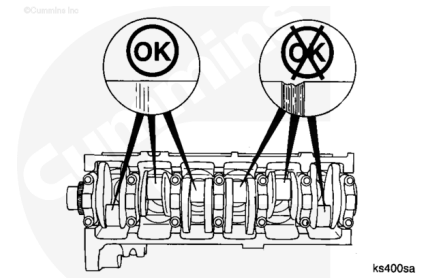
The side of the rod with the cylinder number (bearing tang side) **must** be toward the camshaft.

Rotate the crankshaft until the journal for the rod being installed is at BDC.



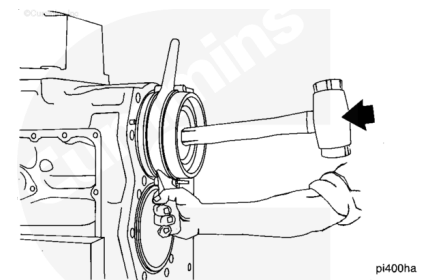
Note : If the engine has a crankshaft with bolt-on counterweights, the journal **must** be at top dead center (TDC).

Check the crankshaft rod journals for damage.

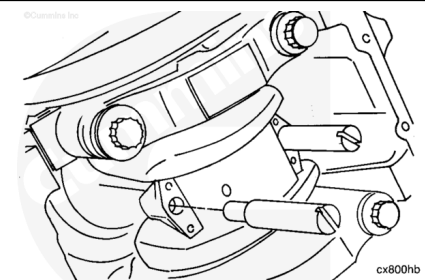


Install the rod and piston until the ring compressor touches the block. Align the rod with the crankshaft journal.

Hold the ring compressor firmly against the block. Use a wooden hammer handle to push the piston into the liner.



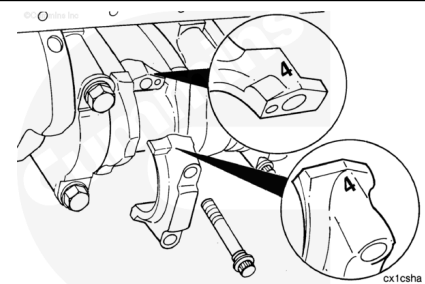
Push the piston into the bore until the rod bearing contacts the crankshaft journal. Remove the guide pins.



The cylinder number on the rod and cap **must** be the same.

The side of the cap with the cylinder number marking (bearing tang side) **must** be toward the camshaft.

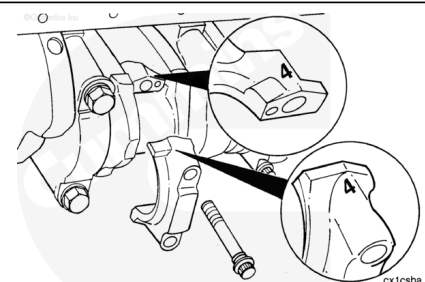
Install the connecting rod cap.



Tighten the capscrews alternately and evenly to pull the cap over the dowel pins. Use the following steps to tighten the capscrews.

Torque Value:

1. 102 n•m [75 ft-lb]
2. 197 n•m [145 ft-lb]
3. 292 n•m [215 ft-lb]



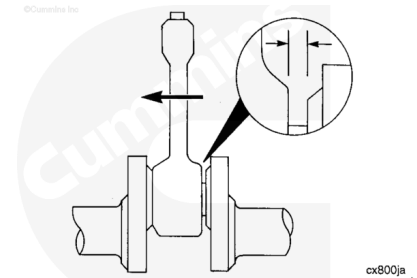
4. 366 n•m [270 ft-lb]
5. Loosen both capscrews to remove all tension.
6. 102 n•m [75 ft-lb]
7. 149 n•m [110 ft-lb]

Plus 90° to achieve correct bolt stretch.

Check the side clearance between the rod and the crankshaft.
The rod **must** move freely from side-to-side.

Measurements over 0.51 mm [0.020 in] **must** be measured with a dial indicator.

If the side clearance is greater than 0.51 mm [0.020 in] do **not** reuse the parts unless the same connecting rod is used on the same crankshaft journal in the same engine. If the parts are reused on the same crankshaft connecting rod journal in the same engine, the maximum side clearance measured with a dial indicator is 5.08 mm [0.200 in]



Connecting Rod and Crankshaft Side Clearance, New or Remanufactured Parts

mm		in
0.20	MIN	0.008
0.35	MAX	0.014

Connecting Rod and Crankshaft Clearance, Used Parts

mm		in
0.51	MAX	0.020

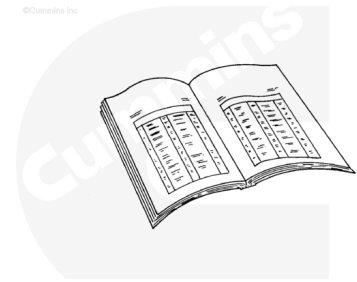
Connecting Rod and Crankshaft Clearance, Same Connecting Rod and Crankshaft Reused on Same Journal of Crankshaft

mm		in
5.08	MAX	0.200

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

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- Install the suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-035-tr.html>)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-025-tr.html>)
- Install the oil pan adapter cover plate. Refer to Procedure 007-026 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-026-tr.html>)
- Replace the cylinder head and related components. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html>)
- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-046-tr.html>)
- Install new oil filter elements. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html>)
- Fill the engine with clean oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)

- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.
- Refer to Procedure 014-002 in Section 14 in (</qs3/pubsys2/xml/en/procedures/20/20-014-002-tr.html>)for engine run in.